Accelerated Quantum Supercomputing: A Hands-On Tutorial on Quantum-Classical Hybrid Workflows Executed on QPUs and GPUs

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I. Abstract

Accelerated quantum supercomputing (AQSC) requires engineers to consider the hardware, software, and infrastructure challenges that arise when integrating quantum processing units (QPUs) into an HPC environment. Fortunately, most of these challenges are hidden from application users, who can leverage these advances to deliver meaningful results. This tutorial will provide attendees with hands-on experience to learn about and utilize this AQSC infrastructure that is being deployed at supercomputing centers around the world.

Participants will gain hands-on experience with the Python API of CUDA-Q. Attendees will have access to GPUs and IQM QPUs hosted at Leibniz Supercomputing Center (LRZ). Live demonstrations and hands-on exercises on how to scale workflows, from a simple `pip install` to deploying multi-GPU, multi-node, and QPU jobs will be given. This will all be tied under 2 key quantum algorithms: Sample-Based Krylov Quantum Diagonalization (SKQD: quantum & HPC) and Generative Quantum Eigensolver (GQE: AI & quantum), both of which have been selected to align with the SC audience and require QPU and GPU computing, emphasizing the need for HPC and AI in quantum computing workflows.

Attendees will also learn about current challenges in scaling quantum hardware and how advancements in quantum error mitigation (QEM) and quantum error correction (QEC) are best suited to address them. They will learn about commonly used QEM and QEC techniques in superconducting devices, execute them in workflows, and perform GPU-accelerated decoding of quantum measurements.

Participants will leave with practical skills in building hybrid applications locally and hands-on experience in scaling them on AQSC infrastructure. Dedicated compute at LRZ for GPUs and IQM QPUs will be provided. Jupyter notebooks, Python files, and detailed instructions on how to connect to the cluster will be made public in a GitHub repository a week before the tutorial which attendees can utilize during and after the tutorial.

Keywords: Quantum Computing, Hybrid Workflows, Quantum Error Correction, Accelerated Quantum Supercomputing, HPC-QC, Quantum Simulation, Quantum Chemistry, Quantum-Classical co-design

II. Description of the tutorial content

A. Goals of the tutorial

The main goal of the tutorial is to familiarise attendees with HPC + QPU environments through interactive demonstrations of applications that leverage this emerging infrastructure. Participants will:

1. Gain hands-on experience with Accelerated Quantum Supercomputing (AQSC)
2. Execute algorithms such as SKQD and GQE on Google Colab via a simple `pip install`, learn how to connect to the LRZ cluster via `SSH`, scale jobs from single GPU to multi-GPU multi-node, and execute on GPUs and QPUs
3. Explore QEM and the basics of QEC, executing codes for superconducting hardware, and performing GPU-accelerated decoding via live demos and hands-on tutorials on IQM's superconducting QPU

B. Relevance for conference attendees

Supercomputing plays a critical role in enabling scalable quantum computing and catalysing breakthroughs by simulating quantum hardware designs, quantum algorithms, and working in conjunction with quantum processing to enhance its performance across the hardware, software, infrastructure and applications stack. This has been explored in depth in recent reviews focused on the use of AI in quantum computing.

Supercomputing centres worldwide are increasing their focus on quantum computing and investing in on-site integrations with physical QPUs. SC 25 and ISC 25 mirrored this trend with an exploding number of talks, panels, papers and workshops dedicated to quantum. Moreover, the vendor showcase floor, which is typically dominated by HPC companies is witnessing an influx of quantum hardware companies.

This tutorial aims to give attendees a hands-on environment to experience the quantum integration at HPC centres they will witness on the vendor showcase floor. Attendees will be better equipped to make greater use of the investments made by supercomputing centres worldwide to advance breakthrough quantum computing research, specifically within the algorithms domain.

We designed the tutorial such that attendees with minimal quantum background can meaningfully participate, while more experienced attendees will appreciate the advanced workflow demonstrations.

C. Target audience

The exercises and algorithms presented in the tutorial are selected particularly to align with the HPC, AI and quantum audience SC attracts. Moreover, the content of each hour of the tutorial builds up competence in a stepwise fashion such that students, early-stage scientists and active researchers in the field can all learn something new from the latest developments in this area.

D. Content level

Beginner: 30% – Overview of AQSC and CUDA-Q, basic quantum programming, supercomputing cluster access and navigation, QPU access.

Intermediate: 40% – Algorithm design (e.g, SKQD and GQE), mid-circuit measurement, streaming input handling, GPU post-processing, basics of QEC, parallelisation techniques, MQSS

Advanced: 30% – execution of GQE workflows on QPUs/GPUs, analysis of noise processes in superconducting QPUs, QEM and QEC with QLDPC codes, and GPU-accelerated decoders

E. Audience prerequisites

- Proficiency in Python programming is required, but no prior experience with CUDA or MPI is needed.
- Basic awareness of quantum computing concepts, such as qubits, quantum gates, and measurements.
- Interest in hybrid quantum-classical computing, scalable quantum applications, and quantum error correction.
- Laptop with the ability to connect to the internet and a terminal window.

F. Tutorial content: half day

Each session includes live demonstrations accompanied by pre-populated Jupyter notebooks (tested on the LRZ cluster and provided via GitHub). Dedicated assistants will be available throughout to support attendees with technical issues. By the end of the tutorial, attendees will have successfully launched and executed jobs on CPUs, GPUs, and superconducting QPU, executed two hybrid quantum-classical algorithms on LRZ, and performed GPU-accelerated QEC decoding on sampled QPU results.
Schedule:

- Overview of Accelerated Quantum Supercomputing (AQSC) - 20 Minutes
- Break - 5 Minutes
- Introduction to AQSC at LRZ - 45 Minutes
- Break - 10 Minutes
- Hybrid Quantum-Classical Algorithms - 45 Minutes
- Break - 10 Minutes
- Superconducting QPUs and Quantum Error - 45 Minutes

G. Cohesion measures

While the presenters are from different institutions, they have multiple ongoing collaborations on quantum education and quantum research projects. More specifically for SC26, the tutorial will be delivered by a tightly coordinated team from NVIDIA, LRZ, UCL and IQM, who have weekly meetings to discuss technical updates on our ongoing research in enhancing the SKQD algorithm presented at this tutorial. A detailed outline was developed jointly, delineating specific sections, learning objectives, and practical exercises assigned to each presenter to ensure a logical progression of topics and maintain consistency. Zohim from NVIDIA has already successfully collaborated on a joint tutorial at SC24 and assisted with the planning of the tutorial at SC25 with NERSC, Infleqtion, and QuEra.

H. Previous editions and updates for SC 26

This tutorial builds upon the successful logistical format and style presented at the recent ISC25 tutorial, however, the content has been updated to ensure attendees learn the latest advances in algorithmic and AQSC innovations. A slightly longer version of this tutorial has been accepted for presentation at ISC26 this June.

The SKQD and GQE algorithm exist in the literature; however the presenters are actively collaborating on a joint research project to introduce various innovations in the workflow. We aim for this to conclude prior to ISC26 so that we can present the latest research in this direction.

Moreover, the active research project we are undertaking is executed at LRZ and involves using the AQSC infrastructure which is only available to internal users. Public access to this has been accelerated due to the needs of our current research project and hence SC attendees will experience a new addition to LRZ's offering which has not been available to the general public before.

IQM has demonstrated quantum error reduction and correction tutorials which are available via IQM Academy. For this workshop, IQM will expand upon existing experiments to reflect its newest developments in 2026 with ongoing work in code discovery, accelerated GPU decoding and AI assisted error mitigation.

These updates reflect the significant advancements expected in AQSC infrastructure, algorithms research and LRZ's quantum-classical integration roadmap entering 2026.

I. Hands-on Feasibility

The content of tutorial notebooks is primarily simple demonstrations of software and workflows already in use by the presenters. The specific format of notebooks distributed to attendees will be tested by each presenter, but will only make use of tools which have been previously tested and published.
Additionally, a version of this tutorial, 4 hours in length, has been accepted for presentation at ISC in June. All tutorials and instructions will follow those presented at ISC, with some alterations in emphasis tailored to the attendees of SC26.

III. Detailed Outline

Topic: Overview of Accelerated Quantum Supercomputing (AQSC) **20 minutes**
Presenter: *Michael Williams de la Bastida (UCL),* ***Assistant:*** *Zohim Chandani (NVIDIA)*
In this section, we will explore challenges posed by quantum computing and how HPC and AI can help overcome them, setting the stage for the technical tutorials that follow. We will also introduce CUDA-Q and its fundamentals.

Topic: Introduction to AQSC at LRZ **45 minutes**
Presenter: *Minh Chung (Leibniz Supercomputing Center)*
In this section, we introduce the HPC–QC infrastructure and integration at the Leibniz Supercomputing Center (LRZ) where CPU-GPU nodes and IQM superconducting QPUs are unified.

Attendees will be made familiar with an introductory quantum computing example (GHZ circuit), chosen so that they can focus on workflow execution and scaling rather than algorithmic complexity. Starting with a `pip install` on Google Colab, they will execute their workflow and understand how this limits scaling to higher qubit counts and why it is essential to move workloads to HPC infrastructure.

We then move onto the LRZ cluster and how to login via `ssh`. Attendees will do hands-on learning of running interactive sessions, managing environments via spack and submitting jobs via Slurm. We introduce the Munich Quantum Software Stack (MQSS) and describe how it enables exascale quantum–classical execution.

Attendees will use all the skills learned above and scale the GHZ circuit example to be executed on multi-node multi-GPU and QPUs. With each step of scale-up we will highlight various strategies for parallelisation with minimal code changes. QPU tasks will use short circuits with rapid turnaround, and GPU-based simulators will be available as fallbacks.

Topic: Hybrid Quantum-Classical Algorithms **45 minutes**
Presenter: *Michael Williams de la Bastida (UCL)*
We realise that attendees at SC have a background in HPC, AI and quantum. To cater to this audience, we picked 2 algorithms: SKQD, which has a quantum and classical HPC-focused postprocessing step and GQE which is a hybrid quantum-classical algorithm using transformers to train quantum circuits.

Integrating quantum algorithms into HPC workflows enables quantum chemistry simulations to scale to molecule sizes relevant for applications in biochemistry, pharmacology, and materials science. In this section, we motivate krylov based methods and how offloading a key step to the QPU to arrive at the SKQD algorithm enables problem sizes previously deemed unfeasible. Attendees will get a flavour of the key computational bottlenecks in the SKQD algorithm, and learn in a hands-on fashion how libraries like CuPy can enable seamless switching between CPU & GPU to get 140x speedup for 22 qubit systems. For attendees with an interest in AI, the generative quantum eigensolver (GQE) leverages a GPT-trained model to generate optimal circuits and overcome fundamental limitations of traditional quantum ML approaches.

Attendees will be walked through each algorithm via Jupyter notebooks and get to execute them on the LRZ infrastructure. Advanced exercises will incorporate parallelisation strategies learned in the previous session to enable scale-up or reduce time to solution.

Topic: Superconducting QPUs and Quantum Error Correction **45 minutes**
Presenter: *Danny Bulmash (IQM)*

Quantum Error Mitigation (QEM) and Quantum Error Correction (QEC) is fundamental in realising practical quantum computing due to the inherent noise in attempting to isolate and precisely control quantum systems. Supercomputing is indispensable in overcoming several bottlenecks in both, including analysing the relationship between physical and logical error rates, designing and optimising QEC strategies, and performing low-latency real-time decoding on quantum hardware.

This section of the tutorial introduces attendees to these topics using various tried and tested tutorials hosted on IQM Academy. Users will be able to follow live demonstrations on how they can prepare logical (error-corrected) qubits for a materials science application on IQM's superconducting quantum computer, run QEC experiments, and use GPUs to accelerate error decoding from data collected by QPUs.

IV. Hands-on feasibility

As directed, this has been described in Section III.

V. Resume of each presenter

Please see the following pages for attached resumees.

VI. Release Agreemnt

We, the authors and presenters of the work contained in this tutorial, agree to release notes relating to this tutorial as a digital copy to attendees of SC26.